The University of Faisalabad (TUF)
Department of Computer Science & Information Technology

SEMESTER PROJECT REPORT

Design and Configuration of a Multi-Campus Network for The University of Faisalabad

Program	BSIT — 6th Semester
Subject	Computer Networks
Tool Used	Cisco Packet Tracer
Campuses	Amin Campus | Saleem Campus
Submitted To	Sir Ibrar
Date	2026

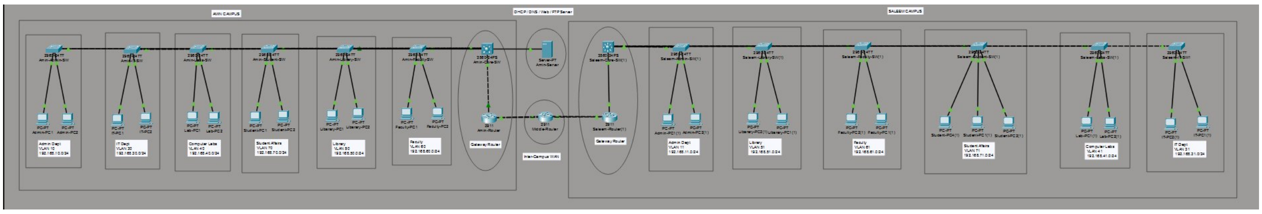
1. Introduction

This report documents the complete design and implementation of a campus network for The University of Faisalabad (TUF) using Cisco Packet Tracer. The network spans two campuses — Amin Campus and Saleem Campus — connected through a dedicated WAN link.

The design follows a hierarchical model: each campus has one gateway router, one Layer 3 core switch, and multiple access-layer switches — one per department. VLANs are used for departmental segmentation, 802.1Q trunking carries tagged traffic between switches and routers, and OSPF provides dynamic routing between campuses. A centralized DHCP server on Amin Campus serves all VLANs on both campuses via ip helper-address.

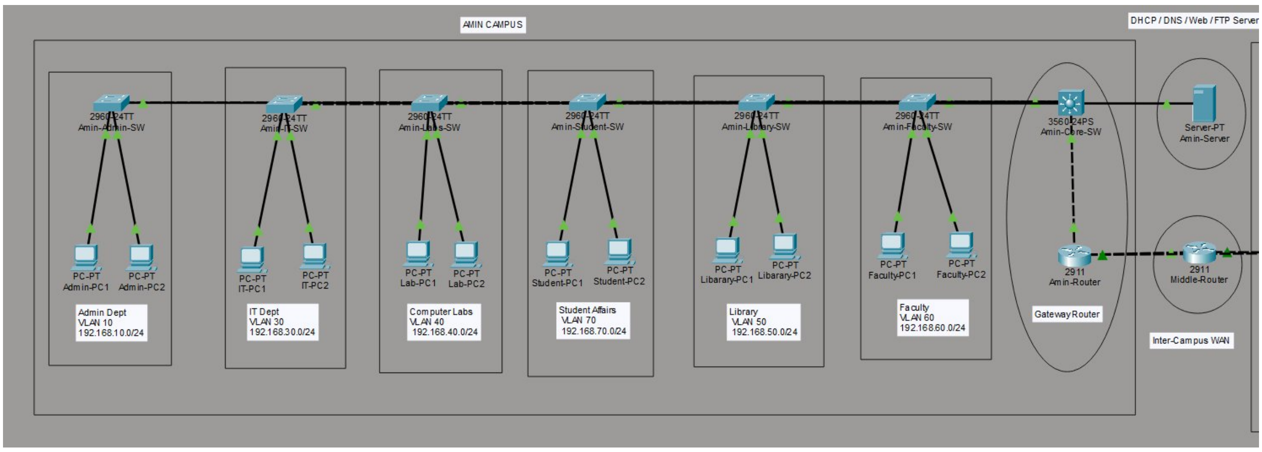
2. Network Topology

The topology below shows both campuses connected via a WAN cloud. Amin Campus sits on the left; Saleem Campus on the right. The centralized DHCP/DNS/Web/FTP server is hosted on Amin Campus.



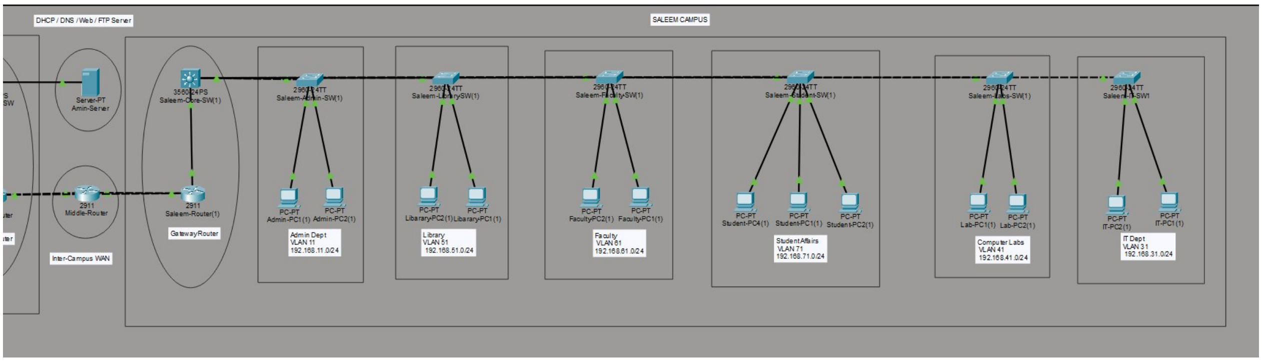
2.1 Amin Campus

Amin Campus consists of Amin-Router (Cisco 2911), Amin-Core-SW (Cisco 3560), six Cisco 2960 access switches (Admin, IT, Labs, Library, Faculty, Students), and Amin-Server hosting DHCP, DNS, Web, and FTP services.



2.2 Saleem Campus

Saleem Campus mirrors the same structure: Saleem-Router (Cisco 2911), Saleem-Core-SW (Cisco 3560), and six Cisco 2960 access switches for the same six departments.


DHCP / DNS / Web / FTP Server
SALEEM CAMPUS
Server-PT
Amin-Server
3560-24PS
2911
Middle-Router
2911
Saleem-Router(1)
GatewayRouter
Inter-Campus WAN
PC-PT
Admin-PC1(1)
PC-PT
Admin-PC2(1)
Admin Dept
VLAN 11
192.168.11.0/24
PC-PT
Libarary-PC2(1)
PC-PT
Libarary-PC1(1)
Library
VLAN 51
192.168.51.0/24
PC-PT
FacultyPC2(1)
PC-PT
Faculty-PC1(1)
Faculty
VLAN 61
192.168.61.0/24
PC-PT
Student-PC4(1)
PC-PT
Student-PC1(1)
PC-PT
Student-PC2(1)
StudentAffairs
VLAN 71
192.168.71.0/24
PC-PT
Lab-PC1(1)
PC-PT
Lab-PC2(1)
Computer Labs
VLAN 41
192.168.41.0/24
PC-PT
IT-PC2(1)
PC-PT
IT-PC1(1)
IT Dept
VLAN 31
192.168.31.0/24

3. VLAN Design

VLANs logically separate departmental traffic across both campuses. Amin Campus uses even-decade IDs (10, 20, 30 …); Saleem Campus uses the next consecutive IDs (11, 31, 41 …).

VLAN ID (Amin / Saleem)	Department	Subnet (Amin)	Subnet (Saleem)
10 / 11	Administration	192.168.10.0/24	192.168.11.0/24
20 / —	Accounts	192.168.20.0/24	—
30 / 31	IT Department	192.168.30.0/24	192.168.31.0/24
40 / 41	Computer Labs	192.168.40.0/24	192.168.41.0/24
50 / 51	Library	192.168.50.0/24	192.168.51.0/24
60 / 61	Faculty	192.168.60.0/24	192.168.61.0/24
70 / 71	Students/Wireless	192.168.70.0/24	192.168.71.0/24
80 / —	Servers (Amin)	192.168.80.0/24	—

4. IP Addressing Scheme

4.1 Amin Campus Hosts

Device	IP Address	Subnet Mask	Default Gateway
Amin-Server	192.168.80.10	255.255.255.0	192.168.80.1
Admin-PC1	192.168.10.x	255.255.255.0	192.168.10.1
Admin-PC2	192.168.10.x	255.255.255.0	192.168.10.1
IT-PC1	192.168.30.x	255.255.255.0	192.168.30.1
IT-PC2	192.168.30.x	255.255.255.0	192.168.30.1
Lab-PC1	192.168.40.x	255.255.255.0	192.168.40.1
Library-PC1	192.168.50.x	255.255.255.0	192.168.50.1
Faculty-PC1	192.168.60.x	255.255.255.0	192.168.60.1
Student-PC1	192.168.70.x	255.255.255.0	192.168.70.1

4.2 Saleem Campus Hosts

Device	IP Address	Subnet Mask	Default Gateway
Admin-PC1(1)	192.168.11.x	255.255.255.0	192.168.11.1
Admin-PC2(1)	192.168.11.x	255.255.255.0	192.168.11.1

IT-PC1(1)	192.168.31.x	255.255.255.0	192.168.31.1
Lab-PC1(1)	192.168.41.x	255.255.255.0	192.168.41.1
Library-PC1(1)	192.168.51.x	255.255.255.0	192.168.51.1
Faculty-PC1(1)	192.168.61.12	255.255.255.0	192.168.61.1
Faculty-PC2(1)	192.168.61.x	255.255.255.0	192.168.61.1
Student-PC1(1)	192.168.71.x	255.255.255.0	192.168.71.1

4.3 WAN Link

Interface	IP Address	Description
Amin-Router Gi0/1	10.10.10.1/30	WAN — Amin Side
Middle-Router Gi0/0	10.10.10.2/30	WAN — Middle
Middle-Router Gi0/1	10.10.10.9/30	WAN — Saleem Side
Saleem-Router Gi0/2	10.10.10.10/30	WAN — Saleem Side

5. Router Configuration

5.1 Inter-VLAN Routing — Sub-Interfaces

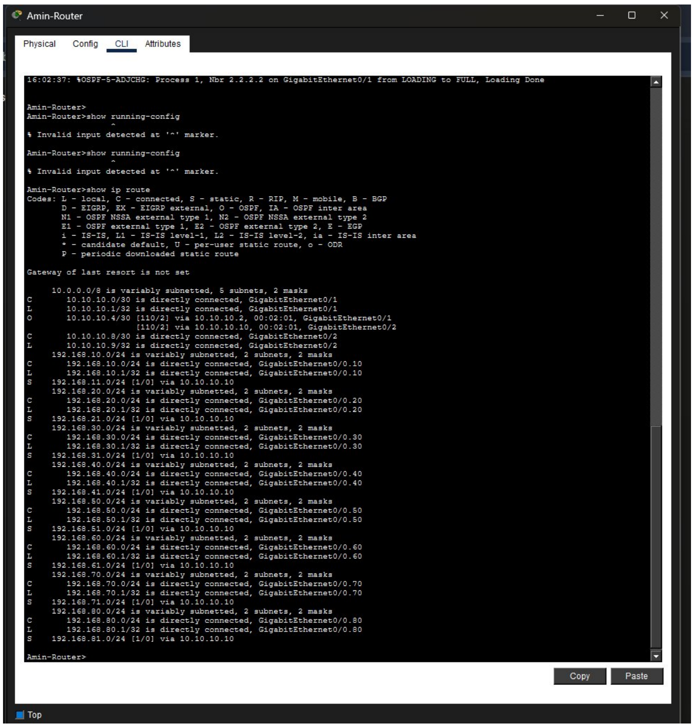
Both routers use router-on-a-stick (802.1Q sub-interfaces on Gi0/0) to route between VLANs. Each sub-interface is assigned the gateway IP for that VLAN. ip helper-address 192.168.80.10 is configured on each Amin-Router sub-interface to relay DHCP requests to Amin-Server.

Sub-Interface	VLAN	IP (Gateway)	ip helper-address
Gi0/0.10	10	192.168.10.1	192.168.80.10
Gi0/0.20	20	192.168.20.1	192.168.80.10
Gi0/0.30	30	192.168.30.1	192.168.80.10
Gi0/0.40	40	192.168.40.1	192.168.80.10
Gi0/0.50	50	192.168.50.1	192.168.80.10
Gi0/0.60	60	192.168.60.1	192.168.80.10
Gi0/0.70	70	192.168.70.1	192.168.80.10
Gi0/0.80	80	192.168.80.1	—

5.2 Saleem-Router CLI — Sub-Interfaces Active

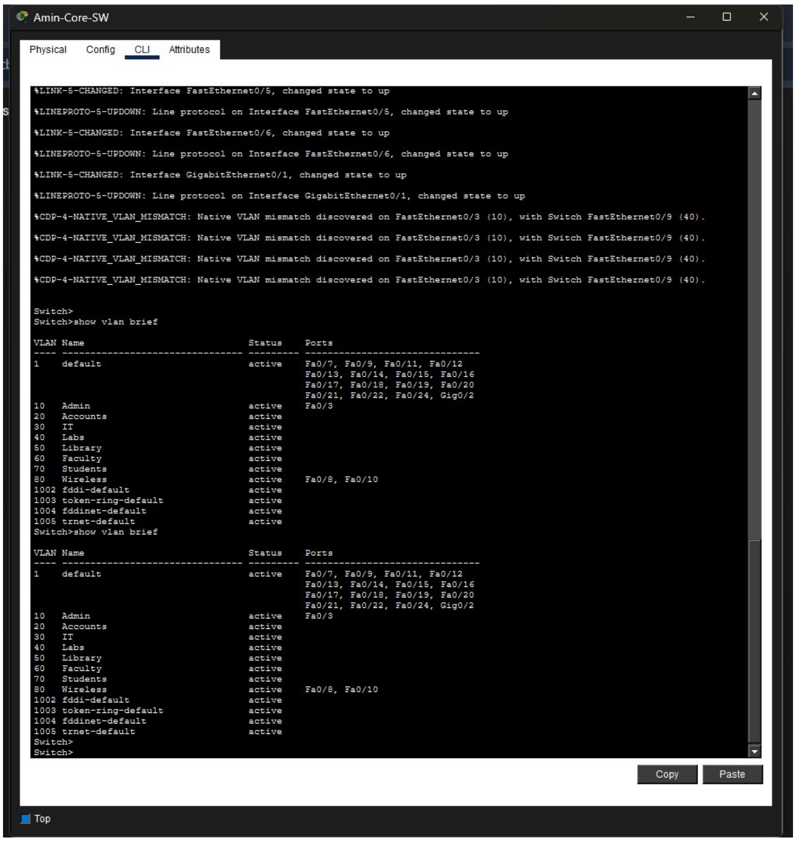
5.3 OSPF Routing

OSPF (Area 0) is configured on Amin-Router, Middle-Router, and Saleem-Router. All VLAN subnets and WAN links are advertised. The show ip ospf neighbor output confirms FULL adjacency between all three routers.



5.4 Amin-Router Routing Table

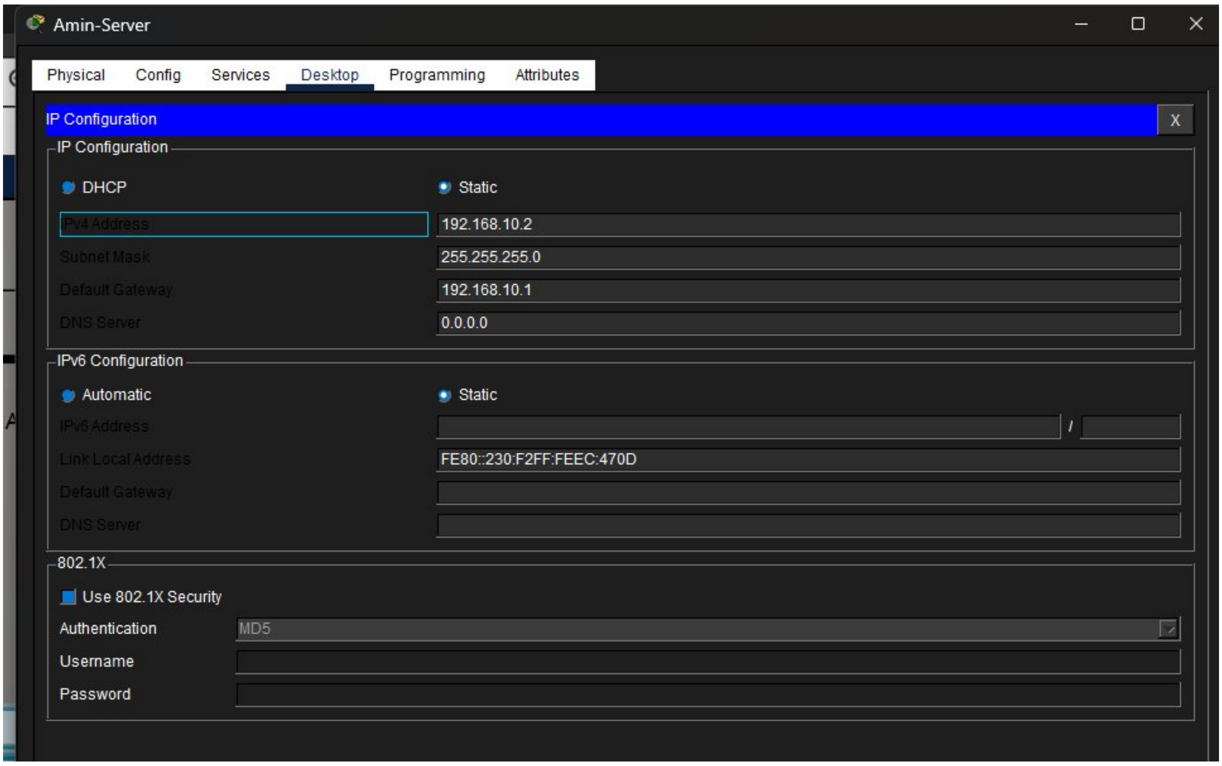
The routing table on Amin-Router shows all directly connected VLANs (C) and all OSPF-learned Saleem Campus subnets (S/O), confirming end-to-end reachability.



6. Switch Configuration

6.1 VLAN Verification — Amin-Core-SW

The show vlan brief command on Amin-Core-SW confirms all VLANs (10, 20, 30, 40, 50, 60, 70, 80) are active and assigned to the correct ports.



6.2 Access Switch Port Assignment

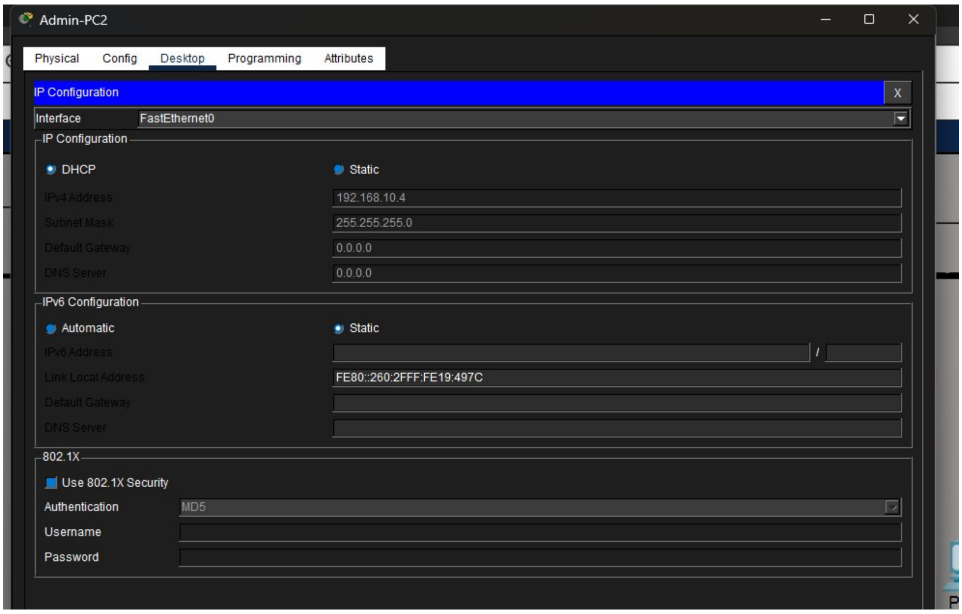
Switch	VLAN	Access Ports	Trunk Port
Amin-Admin-SW	10	Fa0/1–2	Fa0/9 → Core
Amin-IT-SW	30	Fa0/1–2	Fa0/9 → Core
Amin-Labs-SW	40	Fa0/1–2	Fa0/9 → Core
Amin-Library-SW	50	Fa0/1–2	Fa0/9 → Core
Amin-Faculty-SW	60	Fa0/1–2	Fa0/9 → Core
Amin-Student-SW	70	Fa0/1–2	Fa0/9 → Core
Saleem-Admin-SW(1)	11	Fa0/1–2	Fa0/5 → Core
Saleem-IT-SW(1)	31	Fa0/1–2	Fa0/5 → Core
Saleem-Labs-SW(1)	41	Fa0/1–2	Fa0/9 → Core
Saleem-Library-SW(1)	51	Fa0/1–2	Fa0/9 → Core
Saleem-Faculty-SW(1)	61	Fa0/1–2	Fa0/9 → Core

Saleem-Student-SW(1)	71	Fa0/1–2	Fa0/9 → Core

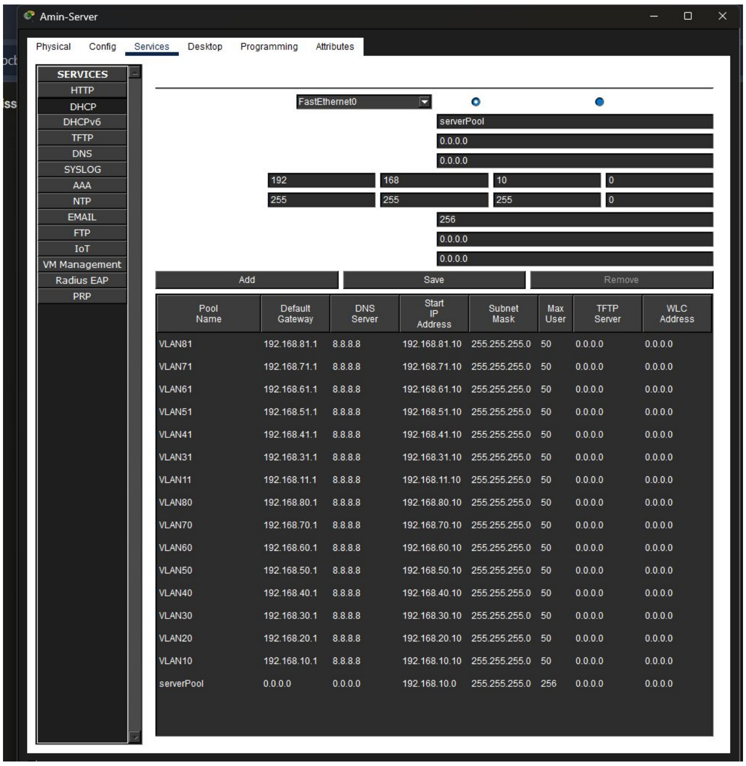
7. DHCP Server Configuration

Amin-Server (192.168.80.10) is configured as the centralized DHCP server for all VLANs across both campuses. Separate DHCP pools are defined for each VLAN with the correct Default Gateway and DNS Server (8.8.8.8).

7.1 Amin-Server — Static IP Configuration



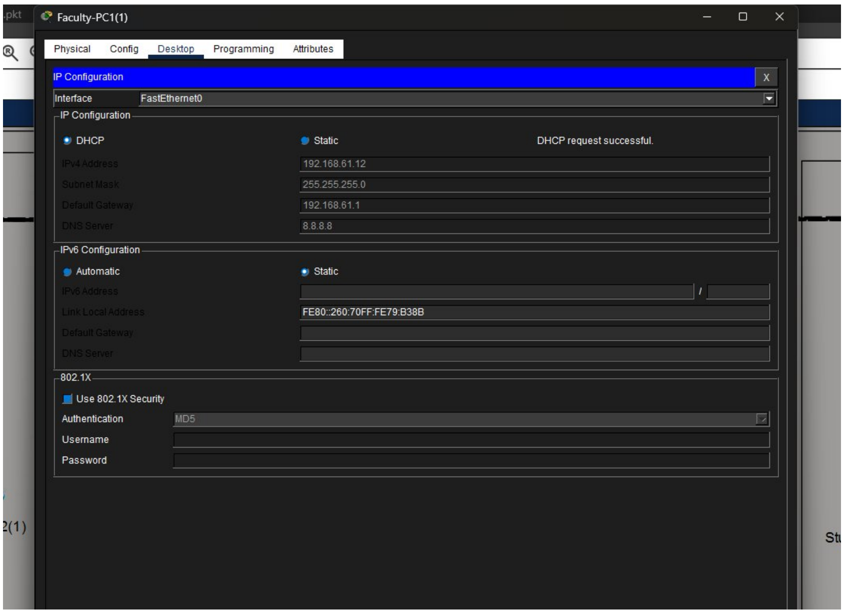
7.2 DHCP Pools — All VLANs



Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
VLAN81	192.168.81.1	8.8.8.8	192.168.81.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN71	192.168.71.1	8.8.8.8	192.168.71.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN61	192.168.61.1	8.8.8.8	192.168.61.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN51	192.168.51.1	8.8.8.8	192.168.51.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN41	192.168.41.1	8.8.8.8	192.168.41.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN31	192.168.31.1	8.8.8.8	192.168.31.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN11	192.168.11.1	8.8.8.8	192.168.11.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN80	192.168.80.1	8.8.8.8	192.168.80.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN70	192.168.70.1	8.8.8.8	192.168.70.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN60	192.168.60.1	8.8.8.8	192.168.60.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN50	192.168.50.1	8.8.8.8	192.168.50.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN40	192.168.40.1	8.8.8.8	192.168.40.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN30	192.168.30.1	8.8.8.8	192.168.30.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN20	192.168.20.1	8.8.8.8	192.168.20.10	255.255.255.0	50	0.0.0.0	0.0.0.0
VLAN10	192.168.10.1	8.8.8.8	192.168.10.10	255.255.255.0	50	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.10.0	255.255.255.0	256	0.0.0.0	0.0.0.0

7.3 DHCP Success — Saleem Campus

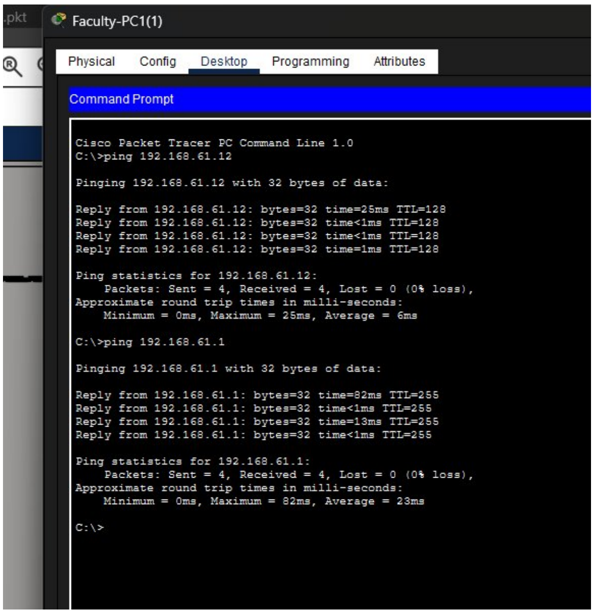
Faculty-PC1(1) on Saleem Campus successfully obtained IP address 192.168.61.12 with Default Gateway 192.168.61.1 from Amin-Server via DHCP relay.



8. Testing and Verification

8.1 Same-VLAN Connectivity Test

Faculty-PC1(1) (192.168.61.12) successfully pinged Faculty-PC2(1) and the default gateway (192.168.61.1) with 4/4 replies and 0% packet loss, confirming intra-VLAN connectivity and correct DHCP gateway assignment on Saleem Campus.



8.2 Test Summary

Test Case	Source	Destination	Result
Same-VLAN Ping	Faculty-PC1(1)	Faculty-PC2(1) 192.168.61.12	PASS — 4/4
Gateway Ping	Faculty-PC1(1)	Gateway 192.168.61.1	PASS — 4/4
OSPF Adjacency	Amin-Router	Saleem-Router	FULL/BDR
WAN Connectivity	10.10.10.1	10.10.10.10	Connected
DHCP Assignment	Saleem Campus PCs	Amin-Server	Successful
Sub-Interfaces	Amin-Router	All VLANs	UP/UP

9. Conclusion

This project successfully demonstrates the design and full configuration of a multi-campus university network for The University of Faisalabad using Cisco Packet Tracer. The network implements industry-standard technologies including VLAN segmentation, 802.1Q trunking, inter-VLAN routing via router-on-a-stick, centralized DHCP with relay, and OSPF dynamic routing.

Both Amin Campus and Saleem Campus are fully operational with seven VLANs each covering all university departments. OSPF adjacency is confirmed in FULL state between all routers. DHCP is centrally managed from Amin-Server and successfully distributing IP addresses across both campuses. Connectivity tests confirm that intra-VLAN communication and gateway reachability are functioning correctly.

The design provides scalability for future expansion, security through VLAN isolation, and simplified management through a centralized server infrastructure.